

CGA: Conditioning as Group Action

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Abstract

Substance: condition a network by *rotating* its features, with angles linear in the condition — then combined conditions compose automatically (rotations add), and the network generalizes to combinations it never saw, where FiLM and far costlier hypernetworks fail. **Significance:** this moves composition from something a network must learn to something the parametrization guarantees — the same move RoPE made for position, here made for conditioning, a mechanism as ubiquitous as position encoding. We present **CGA** (Conditioning as Group Action), a framework that treats feature conditioning $y = T(c)x + \beta(c)$ as a group representation of the condition space: the condition c enters the operator *linearly in an abelian Lie algebra*, $T(c) = P \exp(A(Wc))P^\top$, so the homomorphism law $T(c_1+c_2) = T(c_1)T(c_2)$ holds exactly, for any weights, by construction. Choosing the algebra recovers deployed mechanisms as special cases: the diagonal algebra yields an exactly-compositional variant of FiLM (FiLM’s usual MLP head parametrizes the same orbit *without* the homomorphism, which we show is precisely what fails under composition); the maximal torus $\mathfrak{so}(2)^{d/2}$ with scalar $c=n$ recovers RoPE, and with learned generators the multiplicative GRAPE construction, transplanted from attention positions to arbitrary condition vectors in FiLM’s role. The instantiation we study — planar rotations with angles Wc , conjugated by a Group-and-Shuffle structured orthogonal basis — is exactly orthogonal, identity at $c=0$ and at initialization, and costs $1.1\times$ FiLM’s conditioning FLOPs (a *dense* learned basis provably exceeds any FiLM-comparable budget; at transformer sites the basis can be dropped entirely, as in RoPE). Under a pre-registered, budget-fair protocol against FiLM, conditional LayerNorm, concatenation, and two input-conditioned hypernetworks (up to $48\times$ our parameters), CGA has the lowest compositional OOD error on all five suites we evaluate — two synthetic operator suites and three image-transformation suites on dSprites and 3D Shapes — with complete seed separation throughout ($p=9.1\times 10^{-5}$, $N=10$), including zero-shot extrapolation to compositions longer than any seen in training. Swapping only the linear map for an MLP degrades OOD error by up to 4.3×10^4 ; the most expressive baselines generalize *worst*. One suite’s pre-registered gate returns a negative on a non-OOD criterion — a $1.46\times$ in-distribution fit penalty on 3D Shapes — which we report as registered: exact orthogonality is a robustness–fit trade-off, quantified.

1 The framework

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM [1] computes $y = \gamma(c) \odot x + \beta(c)$; adaLN in DiT blocks [2] is its normalized form; hypernetworks emit $W(c)$ [3]. All are instances of

$$y = T(c)x + \beta(c), \tag{1}$$

differing in the family from which the per-sample operator $T(c)$ is drawn.

Definition 1 (Compositional conditioner). $T : \mathbb{R}^k \rightarrow \text{GL}(d)$ is compositional if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1+c_2) = T(c_1)T(c_2)$ and $T(0) = I$.

Proposition 1 (Exact composition by construction). Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $A : \mathbb{R}^m \rightarrow \mathfrak{a}$ linear, $W \in \mathbb{R}^{m \times k}$, and P orthogonal. Then $T(c) = P \exp(A(Wc)) P^\top$ is a compositional conditioner for any W, P ; if $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.

Proof. $\exp(X)\exp(Y) = \exp(X+Y)$ for commuting X, Y ; $A \circ W$ is linear; conjugation preserves products and $\exp(0) = I$. $\square$

Composition is thus a property of the *parametrization* — it holds at initialization, during training, and off-distribution. Everything learnable (W, P) only selects *which* one-parameter subgroups the condition drives and *how fast*.

Table 1: Compositional OOD error (mean over 10 seeds). CGA is lowest in every column with complete seed separation vs. the strongest hypernetwork baseline ($p=9.1\times 10^{-5}$, $\delta=-1.0$). S1–S3b pass the full pre-registered gate; S4 returns a registered negative on a separate in-distribution criterion (text).

Conditioner	S1	S2	S3	S3b	S4
FiLM	1.3×10^{-2}	1.6×10^{-2}	3.0×10^{-3}	5.9×10^{-3}	4.5×10^{-3}
Concat-MLP	1.5×10^{-2}	1.1×10^{-2}	2.9×10^{-3}	6.3×10^{-3}	5.4×10^{-3}
Cond-LN	2.4×10^{-2}	2.7×10^{-2}	3.2×10^{-3}	6.1×10^{-3}	8.6×10^{-3}
Hypernetwork	1.9×10^{-3}	3.4×10^{-3}	3.8×10^{-3}	7.0×10^{-3}	9.3×10^{-3}
Dyn. low-rank	5.7×10^{-3}	6.1×10^{-3}	5.1×10^{-3}	8.1×10^{-3}	2.0×10^{-2}
MLP-head (abl.)	–	1.1×10^{-3}	2.7×10^{-3}	5.0×10^{-3}	3.3×10^{-3}
Lie (ours)	7.4×10^{-4}	3.3×10^{-8}	1.8×10^{-3}	4.0×10^{-3}	1.2×10^{-3}

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms exactly:

$\mathfrak{a}$	condition	P	recovered mechanism
$\text{diag}(\mathbb{R}^d)$	c	I	<i>exponential FiLM</i> : $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	this work (FiLM’s role)

Standard FiLM parametrizes the same diagonal orbit through an MLP head $\gamma(c)$ — the orbit is right, the homomorphism is absent, and §2 shows empirically that this is exactly what fails under composition: our ablation with an MLP angle head (same orbit, same basis, more compute) loses by up to 4.3×10^4 . The PEFT literature (LoRA, OFT, BOFT, HRA, Group-and-Shuffle [6–10]) owns these operator families as *fixed weight* adaptations; CGA places them in FiLM’s role — per-sample, input-conditioned, activation-space — where the composition question exists.

Cost, and where the basis comes from. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, exp is closed-form planar rotation: $3d$ FLOPs, no matrix exponential. The basis is the entire budget question. A dense learned P costs $4d^2$ FLOPs per sample ($1.52\times$ FiLM’s whole conditioning path at $d=128$) — infeasible under any FiLM-comparable ceiling. Two resolutions: (i) a structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, à la Group-and-Shuffle): total $1.11\times$ FiLM, undercomplete (4,800 parameters vs. $\dim \text{SO}(128)=8,128$) yet sufficient, since P need only expose the planes the condition acts on; (ii) at transformer sites, *no explicit* P : the adjacent dense projections absorb the basis, which is exactly how RoPE deploys. The operator satisfies identity-init ($W=0 \Rightarrow T \equiv I$), $T(0) = I$ structurally, and $\sigma(T(c)) \equiv 1$; all are unit-tested invariants of the release.

2 Experiments

Protocol. Pre-registered throughout: margins, tests ($N=10$ seeds, one-sided Mann–Whitney $p \leq 0.01$, Cliff’s $|\delta| \geq 0.474$), and single-read OOD splits fixed before each run; equal backbones; a shared counter enforces conditioning cost $\leq 1.20\times$ FiLM and parameters $\leq 1.05\times$ the smallest hypernetwork baseline (baselines may be larger — up to $48\times$). Verdicts are reported as registered, including the negative. Baselines: FiLM, conditional LayerNorm, Concat-MLP, dense hypernetwork $I+\Delta W(c)$, dynamic low-rank $I+A(c)B(c)^\top$; plus the MLP-head ablation (identical basis, more compute, no linearity).

Suites. **S1/S2 (synthetic):** learn $y = M(c)x$ for multi-hot compositions of 8 rotation primitives; S2 hides the primitives’ planes behind a random basis B and additionally tests all 56 never-trained *triples*. **S3/S3b (dSprites) and S4 (3D Shapes):** conditional image transformation — encode a real image, apply $T(\Delta)$ on the learned latent for a factor-change Δ , decode; score pixel MSE against the dataset’s unique ground-truth image; OOD = never-trained two-factor change types; S3b/S4 include a *categorical* (shape) factor encoded as a one-hot difference.

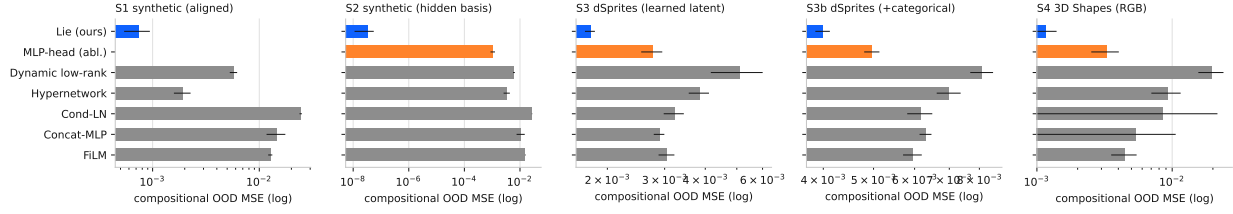


Figure 1: OOD error by conditioner (log scale; blue = CGA, orange = MLP-head ablation).

Findings. (1) **Group-structured conditions: solved, not approximated.** On S2, CGA reaches 3.3×10^{-8} OOD MSE — five orders below the best baseline — and 5.3×10^{-8} on never-trained triples: error does not grow with composition length. Learned angles recover the true generators to 2×10^{-4} rad; a coordinate-aligned ablation ($P=I$) collapses to FiLM’s error, so basis *discovery* carries the result. (2) **The advantage survives learned latents.** On dSprites all arms fit in-distribution identically, isolating composition: CGA’s OOD/in-dist gap is $1.25 \times$ ($1.63 \times$ with the categorical factor) vs. $2.1\text{--}3.3 \times$ for every baseline, a 53.8% (42.9%) margin over the strongest baseline at $39 \times$ fewer conditioning FLOPs. (3) **Linearity is the mechanism.** The MLP-head ablation — same basis, same orbit, more compute — is $4.3 \times 10^4 \times$ worse on synthetic triples and $1.55 \times$ worse on dSprites. (4) **Capacity inverts.** The two hypernetwork arms, the only ones able to represent *any* operator, have the worst compositional gaps on images ($2.7\text{--}26 \times$): expressivity is spent memorizing training condition types. (5) **The registered negative.** On 3D Shapes, CGA again has the best OOD error (87.5% margin; gap $1.13 \times$ vs. $13\text{--}26 \times$) but pays a $1.46 \times$ in-distribution fit penalty that trips the pre-registered no-regression gate ($\leq 1.10 \times$). We report the suite as a negative, as registered: exact orthogonality is a robustness–fit trade-off, and our protocol prices it instead of hiding it.

3 Discussion

CGA does for conditioning what RoPE/GRAPE did for position: it moves a desirable law — composition — from something a network must learn into something the parametrization guarantees. The evidence here is deliberately controlled (64×64 , known factors); the deployment experiment is the adaLN site of diffusion transformers, where the basis-free variant drops in at below-FiLM marginal cost. Two boundaries are already quantified: categorical factors attenuate (but do not remove) the advantage, and richer scenes can price the orthogonality constraint in in-distribution fit — motivating the abelian-algebra generalizations of Proposition 1 beyond the pure torus (e.g., torus $\times$ bounded diagonal) as the next design point. Adjacent prior work learns group actions on latents for world models [11, 12]; CGA differs in living in FiLM’s role under budget parity, and in that composition is exact by construction rather than a training objective.

Reproducibility. All specs pre-registered; every number regenerated from append-only logs; operator invariants and fairness counters are unit tests; amendments and one accounting erratum disclosed in the repository.

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